

DOCUMENTATION: STM32WB FAMILY COMPLETE IC COMPARISON

1. STM32WB FAMILY OVERVIEW

The STM32WB family consists of **over 40 different chip variants** organized into **4 main product lines**:

Product Line	Core Architecture	Target Application	Flash Range	RAM Range
STM32WBx5	Dual-Core (M4 + M0+)	High performance, multi-protocol	256KB - 1MB	128KB - 256KB
STM32WBx0 Value Line	Dual-Core (M4 + M0+)	Cost-effective, balanced features	320KB - 1MB	48KB - 256KB
STM32WB0 Series	Single-Core (M0+)	Ultra-low power, cost-sensitive	192KB - 512KB	24KB - 64KB
STM32WB Modules	Dual-Core (M4 + M0+)	Pre-certified, easy integration	320KB - 1MB	48KB - 256KB

2. COMPLETE IC COMPARISON TABLE

2.1 STM32WBx5 Series (High Performance - Dual Core)

Part Number	Flash	RAM	Package	GPIO	USB	I2C	SPI	UART	BLE	Zigbee
STM32WB55 CGU6	1MB	256KB	UFQFPN -48	30	✓	2	1	2	5.4	✓
STM32WB55 CCU6	256KB	128KB	UFQFPN -48	30	✓	2	1	2	5.4	✓
STM32WB55 CEU6	512KB	256KB	UFQFPN -48	30	✓	2	1	2	5.4	✓
STM32WB55 RGV6	1MB	256KB	VFQFPN -68	49	✓	2	2	2	5.3	✓
STM32WB55 REV6	512KB	256KB	VFQFPN -68	49	✓	2	2	2	5.3	✓
STM32WB55 RCV6	256KB	128KB	VFQFPN -68	49	✓	2	2	2	5.4	✓
STM32WB55 VCQ6	256KB	128KB	UFBGA-129	72	✓	2	2	2	5.3	✓
STM32WB55 VEQ6	512KB	256KB	UFBGA-129	72	✓	2	2	2	5.3	✓
STM32WB55 VGQ6	1MB	256KB	UFBGA-129	72	✓	2	2	2	5.3	✓
STM32WB55 VGY6	1MB	256KB	WLCSP-100	72	✓	2	2	2	5.3	✓
STM32WB55 VEY6	512KB	256KB	WLCSP-100	72	✓	2	2	2	5.3	✓
STM32WB55 VCY6	256KB	128KB	WLCSP-100	72	✓	2	2	2	5.3	✓
STM32WB50 CGU5	1MB	128KB	UFQFPN -48	30	✗	1	1	1	5.4	✓
STM32WB35 CEU6	512KB	96KB	UFQFPN -48	30	✓	2	1	2	5.4	✓
STM32WB35 CCU6	256KB	96KB	UFQFPN -48	30	✓	2	1	2	5.4	✓
STM32WB30 CEU5	512KB	96KB	UFQFPN -48	30	✗	1	1	1	5.4	✓

2.2 STM32WBx0 Value Line (Dual Core - Cost Optimized)

Part Number	Flash	RAM	Package	GPIO	USB	I2C	SPI	UART	BLE	Zigbee
STM32WB15CCU6	320KB	48KB	UFQFPN-48	30	✗	1	1	1	5.3	✗
STM32WB15CCY6	320KB	48KB	WLCSP-49	25	✗	1	1	1	5.3	✗
STM32WB10CCU5	320KB	48KB	UFQFPN-48	30	✗	1	1	1	5.4	✗

2.3 STM32WB0 Series (New - Single Core M0+)

Part Number	Core	Flash	RAM	Package	GPIO	I2C	SPI	UART	BLE
STM32WB09KEV6	M0+	512KB	16KB	VFQFPN-32	20	1	1	1	5.4
STM32WB07CCV6	M0+	256KB	32KB	VFQFPN-48	32	1	2	1	5.4
STM32WB05KZV6	M0+	192KB	24KB	VFQFPN-32	20	1	1	1	5.4
STM32WB06CCV6	M0+	256KB	32KB	VFQFPN-48	32	1	2	1	5.4

2.4 STM32WB Modules (Pre-Certified)

Module	Based On	Flash	RAM	Package	GPIO	USB	Certification	BLE	Price (INR)
STM32WB1MMC	WB15CC	320KB	48KB	LGA-77	27	✗	FCC, CE, IC, MIC	5.4	₹558 – ₹744
STM32WB5MMG	WB55CG	1MB	256KB	LGA-86	49	✓	FCC, CE, IC, MIC	5.4 + Zigbee	₹930 – ₹1116

3. PART NUMBER DECODING

Understanding STM32WB part numbers helps you select the right chip:

Position	Example	Meaning
Family	STM32WB	Wireless MCU family
Series	55, 35, 15, 05	Performance level (higher = more features)
Memory	C, E, G	C=256KB, E=512KB, G=1MB Flash
Package	C, R, V, Y	C=48-pin, R=68-pin, V=100-pin, Y=WLCSP
Temperature	U, V	U=-40 to 85°C, V=-40 to 105°C

Example: STM32WB55CGU6

- 55 = High performance series
- C = 48-pin package
- G = 1MB Flash
- U = -40 to 85°C

4. KEY FEATURE COMPARISON BY SERIES

4.1 Feature Availability Matrix

Feature	WBx5 Series	WBx0 Value Line	WB0 Series	Modules
Dual Core (M4+M0+)	Yes	Yes	No (M0+ only)	Yes
USB Peripheral	Yes (most)	No	No	Varies
LCD Driver	Yes (most)	No	No	Varies
Zigbee/Thread	Yes	No	No	Varies
12-bit ADC	Yes	Yes	Yes	Yes
Quad-SPI	Yes	No	No	Varies
Crypto Engine	Yes	Yes	Yes	Yes

4.2 Package Size Comparison

Package	Size	Pin Count	Used In
UFQFPN-48	7x7mm	48	WB55CG, WB15CC, WB35, WB30
VFQFPN-68	8x8mm	68	WB55RG, WB55RE
UFBGA-129	7x7mm	129	WB55VG, WB55VE
WLCSP-100	4.4x4.0mm	100	WB55VY, WB55VEY
VFQFPN-32	5x5mm	32	WB09, WB05
LGA-77	6.5x10mm	77	WB1MMC (module)
LGA-86	7.3x11mm	86	WB5MMG (module)

5. PRICE COMPARISON (1-unit pricing)

Part Number	Flash	Key Feature
STM32WB09KEV6	512KB	Lowest cost, single-core
STM32WB07CCV6	256KB	Good value, single-core
STM32WB30CEU5	512KB	Value line, dual-core
STM32WB15CCU6	320KB	USB + Zigbee
STM32WB35CCU6	256KB	USB + Zigbee
STM32WB55CGU6	1MB	Your dev board chip
STM32WB1MMC	320KB	Pre-certified module
STM32WB5MMG	1MB	Full module

6. COMPLETE LIST OF ALL STM32WB VARIANTS

Based on STMicroelectronics official documentation and distributor data:

Modules (2 variants) (Ready-to-use, no RF design)

- STM32WB1MMC, STM32WB5MMG

WBx5 Series (16+ variants) (High performance, full wireless support)

- STM32WB55CG, STM32WB55CC, STM32WB55CE
- STM32WB55RG, STM32WB55RC, STM32WB55RE
- STM32WB55VG, STM32WB55VC, STM32WB55VE
- STM32WB55VY, STM32WB55VEY, STM32WB55VCY
- STM32WB50CG
- STM32WB35CE, STM32WB35CC
- STM32WB30CE

WBx0 Value Line (3+ variants) (Low cost, BLE only)

- STM32WB15CC, STM32WB15CCY
- STM32WB10CC

WB0 Series (11+ variants) (Ultra low cost, basic IoT)

- STM32WB09KE, STM32WB09TE
- STM32WB07CC, STM32WB07KC
- STM32WB06CC, STM32WB06KC
- STM32WB05KZ, STM32WB05TZ

TOTAL: 32+ distinct STM32WB variants

7. ADC QUALITY ACROSS ALL STM32WB BOARDS

All STM32WB series chips share the **same high-quality 12-bit ADC**. There is no difference in ADC quality between the WB55, WB15, or WB0 series.

7.1 ADC Specifications (Common to All STM32WB Chips)

ADC Feature	Specification	Why It Matters for Your Project
Resolution	12-bit native	Reads 0-4095 values - excellent for sensors
Max with Oversampling	Up to 16-bit	Software averaging for noise reduction
Max Conversion Rate	4.26 Msps	Very fast - reads sensor instantly
Min Sampling Time	39 ns	Can capture fast-changing signals
External Channels	Up to 16	Connect multiple sensors
Internal Channels	3 (Vref, temperature, battery)	Monitor internal health
Low Power Mode	200 μ A/Msps	Battery friendly
Single-ended Mode	Yes	Simple sensor connection
Differential Mode	Yes	Better noise rejection
Hardware Oversampling	Yes	Built-in averaging

8. CERTIFICATIONS ACROSS ALL STM32WB BOARDS

All STM32WB chips share the **same certifications** at the silicon level.

8.1 Common Certifications

Certification	Region	Status	What It Means for Your Product
ETSI EN 300 328	Europe	Compliant	Legal to sell in Europe
EN 300 440	Europe	Compliant	Short-range radio legal
FCC CFR47 Part 15	USA	Compliant	Legal to sell in USA
ARIB STD-T66	Japan	Compliant	Legal to sell in Japan
RoHS 2011/65/EU	Europe	Compliant	No hazardous materials
RoHS 2015/863	Europe	Compliant	Extended chemical ban
REACH	Europe	Unaffected	Safe chemicals
Bluetooth SIG	Global	Certified (BLE 5.2/5.3/5.4)	Official Bluetooth logo allowed

8.2 Certification by Series

Certification	WBx5 Series	WBx0 Series	WB0 Series
RF (ETSI, FCC, ARIB)	Yes	Yes	Yes
RoHS / REACH	Yes	Yes	Yes
Bluetooth SIG	BLE 5.3/5.4	BLE 5.2	BLE 5.4
Zigbee / Thread	Yes	No	No

9. RECOMMENDATION FOR NEW DESIGN

9.1 Is the STM32WB Family Recommended for New Designs?

YES - The STM32WB family is recommended for new designs.

STMicroelectronics has committed to a **10-year longevity plan** for the STM32WB series, ensuring long-term availability for your product. The family remains **Active** (in volume production) with ongoing support

9.2 Which Chips to Choose (Recommended)

Recommendation	Chip / Series	Why
BEST CHOICE	STM32WB55 Series (WB55CEU6, WB55CGU6)	Full features, dual-core, USB, Zigbee, most mature, best documentation
GOOD CHOICE	STM32WB15 Series (WB15CCU6 - Rev Z)	Lower cost, dual-core, no USB (good if you removed USB-C port)
EASIEST CHOICE	STM32WB1MMC Module	Pre-certified, no RF design needed, saves 6-12 months development time

10. FINAL SELECTION OF BOARD & CHIP

Based on your complete project requirements (ADC reading + BLE parameter updates + GUI communication via UART header + Headers-only + Compact design + Battery-powered), here is the **final selection** with proper reasoning.

10.1 Final Chip Selection – Complete Specifications

Parameter	Value
Part Number	STM32WB55CEU6
Series	STM32WBx5 (High Performance)
Package	UFQFPN-48 (7mm x 7mm)
Core	Dual-Core: Cortex-M4 (64 MHz) + Cortex-M0+ (32 MHz)
Flash	512 KB
RAM	256 KB
Operating Voltage	1.71V – 3.6V
Temperature Range	-40°C to +85°C
Status for New Design	FULLY RECOMMENDED

Wireless (BLE)

Parameter	Value
BLE Version	5.4
Bluetooth SIG Certification	✓ Certified
Multi-Protocol	BLE + Zigbee + Thread (IEEE 802.15.4)
RX Sensitivity	-96 dBm (BLE 1Mbps)
TX Power	Up to +6 dBm (1 dB steps)
Data Rate	Supports 2 Mbps
Integrated Balun	✓ Yes

Communication Interfaces

Interface	Count	Pins Available	Purpose for Your Project
USART	1	PA9 (TX), PA10 (RX)	GUI Communication via UART Header
LPUART	1	PA2 (TX), PA3 (RX)	Low-power wake-up from BLE
I2C	2	PB6 (SCL), PB7 (SDA) – I2C1 PB10 (SCL), PB11 (SDA) – I2C2	External sensors (optional)
SPI	2	PA5, PA6, PA7 – SPI1 PB13, PB14, PB15 – SPI2	High-speed devices (optional)
USB 2.0 FS	1	PA11 (D+), PA12 (D-)	Optional – you are NOT using (headers-only)
SWD	1	PA13 (SWDIO), PA14 (SWCLK)	Debugging & Programming
GPIO	30	Various	General purpose I/O

ADC (Analog to Digital Converter)

Parameter	Value
Resolution	12-bit native
Max with Oversampling	Up to 16-bit
Max Conversion Rate	4.26 Msps
Number of Channels	Up to 16 external channels
Internal Channels	3 (Vref, temperature sensor, battery)
Low Power Mode	200 μ A/Msps
Single-ended Mode	Yes
Differential Mode	Yes

Power Consumption

Mode	Current	When to Use
Shutdown	13 nA	Long-term storage
Standby (with RTC)	600 nA	Deep sleep, wake on timer
Stop	2.1 μ A	Light sleep, fast wake
Active (CPU)	53 μ A/MHz	Processing (ADC, GUI logic)
BLE Receive	4.5 mA	Listening for commands
BLE Transmit @0dBm	5.2 mA	Sending data

Security Features

Feature	Description
AES Engine	256-bit hardware encryption
PKA	Public Key Accelerator
RNG	True Random Number Generator
PCROP	Proprietary Code Readout Protection

Certifications

Certification	Region	Status
ETSI EN 300 328	Europe	Compliant
EN 300 440	Europe	Compliant
FCC CFR47 Part 15	USA	Compliant
ARIB STD-T66	Japan	Compliant
RoHS 2011/65/EU	Europe	Compliant
REACH	Europe	Compliant
Bluetooth SIG	Global	Certified (BLE 5.4)

Pin Summary for Your Headers-Only Design

Header	Pin	STM32 Pin	Function	Connect To
SWD (4-pin)	1	PA13	SWDIO	ST-Link SWDIO
	2	PA14	SWCLK	ST-Link SWCLK
	3	NRST	Reset	ST-Link NRST
	4	GND	Ground	ST-Link GND
UART (3-pin)	1	PA9	USART1_TX	USB-to-Serial RX
	2	PA10	USART1_RX	USB-to-Serial TX
	3	GND	Ground	USB-to-Serial GND
Power (2-pin)	1	VDD (3.3V)	Power input	3.3V Regulator
	2	GND	Ground	Ground
ADC (1-pin)	1	PA0 (ADC1_IN0)	Analog Input	Sensor Output

I2C (Optional)	1	PB6	I2C1_SCL	External Sensor SCL
	2	PB7	I2C1_SDA	External Sensor SDA
	3	GND	Ground	Ground

Summary

Category	Specification
Chip	STM32WB55CEU6
BLE Version	5.4
ADC	12-bit, 4.26 Msps, 16 channels
GUI Interface	UART (PA9/TX, PA10/RX)
Debug Interface	SWD (PA13, PA14, NRST)
I2C Available	2 channels (PB6/PB7, PB10/PB11)
UART Available	1x USART + 1x LPUART
GPIO Total	30
Standby Power	600 nA
Active Power	53 μ A/MHz
Package	UFQFPN-48 (7x7mm)
New Design	RECOMMENDED

10.2 Point Reasons for Selection

	Reason	Explanation
1	Dual-Core Architecture	BLE stack runs on dedicated M0+ core. Your ADC reading and GUI communication run on M4 core. No interference between wireless and application tasks.
2	12-bit ADC (4.26 Msps)	Excellent for sensor reading. Hardware oversampling up to 16-bit. Same high-quality ADC across all STM32WB series.
3	BLE 5.4 Certified	Latest Bluetooth standard. Supports parameter updates over BLE. Bluetooth SIG certified.
4	Perfect Memory Size	512KB Flash / 256KB RAM. Not too little (like WB15's 48KB RAM) and not wasteful overkill (like WB55's 1MB Flash). Perfect balance for your needs.
5	UART for GUI Communication	USART1 on PA9 (TX) and PA10 (RX) – perfect for connecting USB-to-Serial converter for GUI communication.
6	SWD for Debugging	PA13 (SWDIO) and PA14 (SWCLK) – standard debug interface. Works with ST-Link V2/V3.
7	Ultra-Low Power	600 nA standby, 13 nA shutdown, 53 μ A/MHz active. Excellent for battery-powered products.
8	Mature & Stable	Released in 2019, all bugs fixed, no major errata. Safe for production.
9	10-Year Longevity	ST guarantees availability until at least 2036. Future-proof for your product.
10	All Certifications Included	FCC, CE, ARIB, RoHS, REACH, Bluetooth SIG. Saves certification cost and time.

10.3 Why NOT Other Chips

Chip	Why NOT Selected
STM32WB55CGU6 (1MB Flash)	Too expensive, too much memory – overkill for your needs
STM32WB15CCU6	Less memory (320KB Flash / 48KB RAM) – tight for future; has errata on Revision B
STM32WB09KE (WB0 Series)	Single-core – BLE and your code share one processor; NOT recommended
nRF52840 (Nordic)	More expensive, new SDK to learn
ESP32-C3	Higher power consumption, has WiFi you don't need